Business Requirements Document

SignFlow — Digital Document Approval Workflow for a University

Case Study no. 89
Course: Software Engineering & Project Management, Semester III
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1. Executive summary

University administration currently moves about 2,000 approval files each month between offices. The paper process takes an average of 12 days, and files can be lost while in transit. SignFlow is proposed as a controlled digital workflow for submitting, routing, approving, signing, reminding, and tracking these files.

The recommended first release covers purchase requests, leave, and event approvals. Together they represent 1,600 files per month, or 80% of the stated monthly volume. The supplied per-file-type effort gives an indicative 27 person-days for this three-type option, compared with 108 person-days for all 12 file types.

2. Business problem

Current condition	Business effect
Files move physically between offices	Files can be delayed or lost
Average approval time is 12 days	Administration cannot meet a faster service target
Status is difficult to see while a file is moving	Staff spend time asking for updates
Senior officers have mixed comfort with computers	A digital process may be rejected or bypassed without assisted adoption
Signature validity must be established	A technically working process may still be unusable for official approvals

3. Business objectives and success measures

Objective	Measure	Target
Reduce approval delay	Average approval time	Under 4 days
Prevent lost records	Submitted files that cannot be retrieved by ID	Zero
Cover the highest-volume work first	Monthly volume handled by first release	1,600 of 2,000 files, or 80%
Preserve accountability	Test transactions with complete audit records	100%
Make digital signing acceptable	Production signature method	Written legal/compliance approval before release
Make progress visible	Reporting cadence	After each phase and after each pilot reporting period

4. Stakeholder needs

Stakeholder	Need	Decision or contribution
Registrar	Faster, traceable approvals and a clear exception path	Owns scope, acceptance, and escalation
IT head	A maintainable workflow with measurable quality	Owns technical design and release recommendation
Administration staff	Simple intake, clear errors, and status visibility	Supplies process knowledge and file metadata
Section officers	Correct routes and manageable inboxes	Confirms rules and performs UAT
Senior officers	A short approval path and confidence in the signature step	Participates in walkthroughs and UAT
Records/audit staff	Complete, searchable history	Validates retention and audit evidence
Legal/compliance representative	Signature method that can be accepted for official records	Approves the method in writing

5. Business scope

In scope for the proposed first release

- Digital registration of purchase requests, leave, and event approvals.
- Routing by file type and approved amount bands.
- Inbox views for reviewers and approvers.
- Approval, rejection, return, and signature actions.
- Reminders, escalation, search, audit history, and reporting.
- Assisted onboarding for senior officers who need it.

Later release or controlled interim process

The remaining file types should be handled through a documented interim process until their rules are baselined. The all-types option remains a valid later release and is estimated at 108 person-days using the supplied figures.

Outside this business case

SignFlow does not replace the university’s finance, HR, or event systems; define a new legal framework; decide paper-disposal policy; or procure hardware and network infrastructure.

6. Business process: proposed future state

1. An authorised staff member creates a file and attaches the required documents.
2. SignFlow validates required information and issues a unique file ID.
3. The routing rules select the approval path using file type and amount where applicable.
4. The current owner receives the file in an inbox and sees the due date.
5. The approver approves, rejects, or returns the file with a reason.
6. If required, the signature service returns a validation result.
7. SignFlow records each event, sends reminders, and escalates overdue work.
8. The requester and authorised managers can see status and reports.

7. Business rules

Rule	Business requirement
BR-01	Every submitted file receives a unique identifier.
BR-02	A file cannot be routed until required metadata and attachments pass validation.
BR-03	Routing is determined from approved file-type and amount rules.
BR-04	An approval requiring a signature cannot become final until the signature result is validated.
BR-05	A return or rejection requires a reason.
BR-06	Every lifecycle event is recorded in an append-only audit history.
BR-07	An overdue item creates the configured reminder or escalation event.
BR-08	The production signature method requires written legal/compliance approval.

8. Business requirements catalogue

ID	Requirement	Priority	Linked SRS
BR-01	Administration staff shall register and track approval files digitally.	Must	FR-01, FR-04
BR-02	The process shall cover the top three monthly file types first.	Must	Scope, FR-03
BR-03	The process shall reduce the average approval time below 4 days.	Must	NFR-05, FR-10
BR-04	No submitted file shall be lost from the digital record.	Must	NFR-06, FR-08
BR-05	Approvals and signatures shall be attributable and auditable.	Must	FR-05, FR-06, FR-08, NFR-01, NFR-02
BR-06	Users shall receive reminders and managers shall see overdue work.	Must	FR-07, FR-10
BR-07	Senior officers shall have an assisted path during adoption.	Should	Stakeholder need; UI design
BR-08	The design shall allow later expansion to all 12 file types.	Should	Architecture and project plan

9. Options considered

Option	Coverage	Supplied effort	Business view
Digitise all 12 types immediately	2,000 files, assuming the listed categories account for the full volume	108 person-days	Broader first release, but more rules and more change at once
Start with top three	1,600 files / 80%	27 person-days	Recommended pilot; leaves a controlled 20% interim process

10. Risks and constraints

The main business risks are resistance from senior officers, lack of legal acceptance for signatures, incorrect routing, incomplete rules for less common types, and continued delay if queue ownership is unclear. The project risk register contains the scored risks, owners, and responses.

11. Business acceptance

The Registrar accepts the business case when the selected release boundary is recorded, the target measures and owners are agreed, the signature method has a legal/compliance review path, and the interim process for the remaining file types is documented.

Registrar: _____
IT head: _____
Date: _____